

Element A

Problem Identification and Justification

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I. Problem Statement

Professionals in the medical field lack a fast and risk-free method for physical experience, essential for surgery training to be more efficient with patient care. Medicare currently pays \$180,000 per medical resident each year, a total of over 25 billion dollars,¹ and a great proportion of that is spent on the acquisition and usage of cadavers, a lengthy process with ethical and practical issues.

II. Justification

The lack of more effective and efficient training in high-risk professions has several legitimate problems, such as costs, patient safety, and technological advancements.

A. Overview

Medical education has long relied on traditional methods, such as hands-on practice with cadavers, to bridge the gap between theoretical knowledge and practical application. These methods, at most medical institutions, are proven as the best way to actively train for surgeries and post-mortems. However, these methods face significant limitations. Cadavers are rarely donated to medical institutions, and many fail to meet strict requirements, creating a worldwide shortage. Additionally, the unregulated nature of the cadaver field yields great corruption, raising ethical debates for families involved. As the demand for skilled healthcare professionals continues to rise, these challenges threaten the quality and accessibility of medical training.

This report explores the critical role of physical learning in medical education, focusing on the limitations of cadaver-based training, the escalating costs of conducting said training impacting the costs of a medical degree, and how these negatives affect the field.

¹“The Costs of Training Internal Medicine Residents in the United States.” The American Journal of Medicine, U.S. National Library of Medicine, pubmed.ncbi.nlm.nih.gov/25063649/. Accessed 12 Nov. 2024.

B. Physical Limitations

In terms of education and refinement, real-world knowledge application is the best way to solidify one's skills. This is especially true in the medical field, which requires physical education to retain information and be more efficient in the surgery bed. Information is better integrated by having the body perform related actions, creating neural patterns to grasp concepts and ideas.

Performance vs. perception

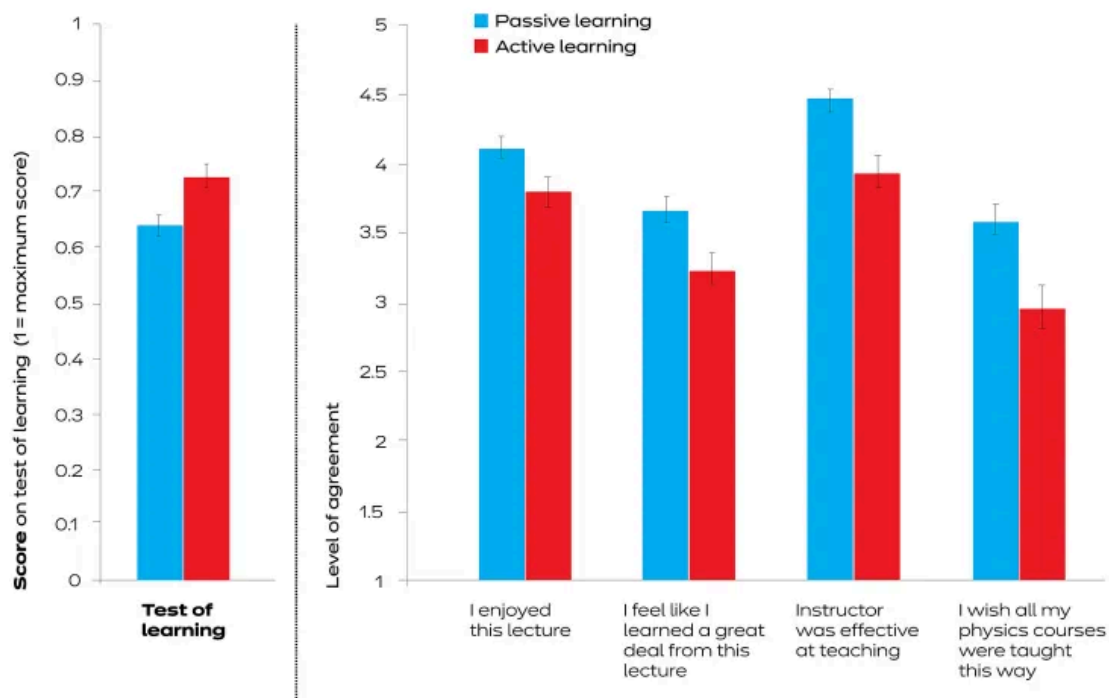


Figure 1.1²: “Measuring actual learning versus the feeling of learning in response to being actively engaged in the classroom,” Louis Deslauriers, Logan S. McCarty, Kelly Miller, Kristina Callaghan, and Greg Kestin

Figure 1.1 shows the impact of passive vs. active learning. Passive learning refers to the general techniques of learning through lectures and textbook reading. Although many felt more content with passive learning techniques (on average reporting ~0.5 levels of engagement more for passive than active), most performed better with active learning techniques.

In terms of active learning, the medical field utilizes various practices, mainly through the use of cadavers to practice surgeries and post-mortems/autopsies. Many institutions hold cadavers as a staple in every medical student's education – able to get hands-on experience with surgical procedures, visualizations of organs and biological structures, and experience with the environment of surgery.³

² Deslauriers, Louis, et al. *Proceedings of the National Academy of Sciences*, vol. 116, no. 39, 4 Sept. 2019, pp. 19251–19257, doi:10.1073/pnas.1821936116.

³Aspiring Docs. “What It's Like to Take Anatomy Lab | Students & Residents.” Students Residents - AAMC, 2021, <https://students-residents.aamc.org/aspiring-docs-fact-sheets-what-medical-school/what-its-take-anatomy-lab>. Accessed 13 November 2024.

However, these can have many limitations due to the nature of the resource itself; cadavers are rare and valuable. Many countries across the globe face shortages of cadavers for medical training purposes. Turkey is an example: one Ege University reported having enough resources for only 15 to 20 students out of their 400 first-year students. The country has low donations, receiving only five to six from their most populous areas, and less elsewhere.⁴ In more wealthy countries such as the UK, only 1,300 corpses are accepted and used in medical schools across all institutions, which is inadequate for the demand for new medical students.⁵ A large part of the shortage is the fault of inefficiencies in the cadaver process.

Medical cadavers endure a long tedious process of consent and screening before being sent to institutions across the country. In the US, the Uniform Anatomical Gift Act (UAGA), enacted in 1968, requires consent from the person donating their body or from a decedent or close relative.⁶ Then, multiple medical screenings are made to dictate whether the cadaver is fit for study: must be free of transmissible diseases such as hepatitis or HIV, have undergone recent surgery, or have an abnormal body mass index (BMI). Only then can the body be transported safely to the various medical institutions across the U.S.⁷ With many families denying consent for these medical causes, having bodies that pass the strict criteria is a rare combination in this dwindling supply of cadavers; only 20,000 bodies are donated yearly⁸, and a mere proportion of those are put towards educating the ever-growing body of medical students.

Additionally, the medical body donor system raises many ethical and legal debates. Although legislation like the UAGA protects various rights in the field, legal definitions and regulations are largely open-ended, leaving uses for donations to interpretation. This ambiguity surrounding consent, distribution, and purposes of cadavers leads to various ethical dilemmas.

⁴ “Medical faculties rely on cadaver imports amid severe shortage - Türkiye News.” *Hürriyet Daily News*, 7 November 2024, <https://www.hurriyetdailynews.com/medical-faculties-rely-on-cadaver-imports-amid-severe-shortage-202352>. Accessed 13 November 2024.

⁵ Kleeman, Jenny. “The great cadaver shortage: inside doctors' latest crisis.” *The Times*, 28 March 2024, <https://www.thetimes.com/life-style/health-fitness/article/doctors-latest-crisis-a-chronic-shortage-of-cadavers-to-train-on-tbrtm9jfs>. Accessed 13 November 2024.

⁶ Wex Definitions Team. “Uniform Anatomical Gift Act | Wex | US Law | LII / Legal Information Institute.” *Law.Cornell.Edu*, 2021, https://www.law.cornell.edu/wex/uniform_anatomical_gift_act. Accessed 13 November 2024.

⁷ Science Care. “How body donation to science works.” *Science Care*, <https://www.sciencecare.com/how-does-the-body-donation-process-work>. Accessed 13 November 2024.

⁸ Sherman, Justin. “Inside the largely unregulated market for bodies donated to science: 'It's harder to sell hot dogs on a cart.'” *CBS News*, 23 March 2023, <https://www.cbsnews.com/news/bodies-donated-to-science-largely-unregulated-cbs-reports/>. Accessed 17 November 2024.

For instance, cadavers may be important for research in other fields such as military, automotive, and other non-medical projects such as crime-scene investigations, vehicular safety, or weapons testing. Although the UAGA protects consent, the breach of contracts, “the misleading of families across the industry”, is common. Companies sell bodies to organizations without notice. This prompts more misuse due to its profitability. Tissue banks constantly trade in underground networks to criminals and other fields that require but don’t get bodies legally. These actions by a great proportion of the cadaver business then cause turmoil and protest against the system, creating a stigma of danger and uncertainty. Many are more reluctant to donate their loved ones’ bodies due to this stigma.⁹

Overall, as proven, the best practice to learn is through physical experience. Cadavers are an important part of a medical student’s education,¹⁰ but pose various issues such as a limited supply and various ethical dilemmas. VR technology could aid in this dilemma: more accessible, easier to produce on a large scale, and potentially cheaper.

⁹ Sherman, Justin. “Inside the largely unregulated market for bodies donated to science: ‘It’s harder to sell hot dogs on a cart.’” CBS News, 23 March 2023, <https://www.cbsnews.com/news/bodies-donated-to-science-largely-unregulated-cbs-reports/>. Accessed 17 November 2024.

¹⁰ Aspiring Docs. “What It’s Like to Take Anatomy Lab | Students & Residents.” Students Residents - AAMC, 2021, <https://students-residents.aamc.org/aspiring-docs-fact-sheets-what-medical-school/what-its-take-anatomy-lab>. Accessed 13 November 2024.

C. Costs

In 2024, the average cost of obtaining a medical degree is approximately \$238,420. A significant portion of this cost goes toward training programs involving physical resources and specialized equipment.

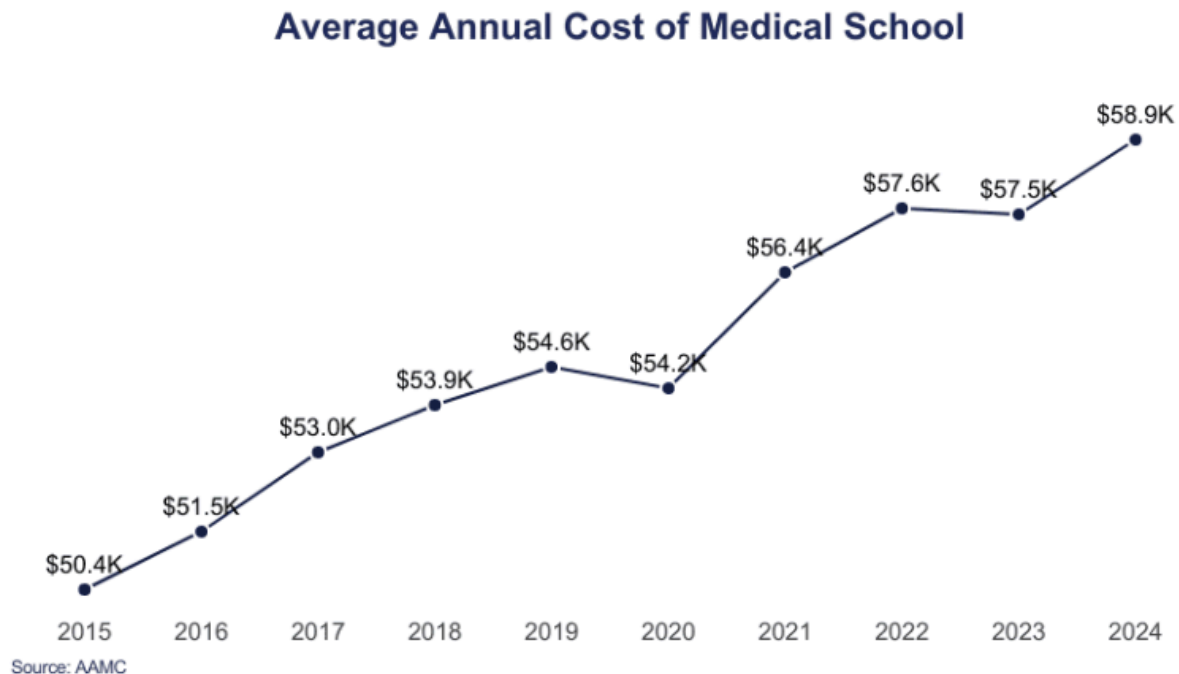


Figure 1.2¹¹: Average annual costs of medical school per year

Figure 1.2 shows the cost of one year of medical school in 2024, which has grown by more than 8.5% since 2020. Cadavers and study models are used widely throughout medical schools, which are expensive to procure, prepare, and store, explaining this trend of increasing cost. A single cadaver can cost between \$2,000 and \$3,000, excluding preservation and handling costs.¹² There are also special surgical training equipment like advanced surgical beds and tools for practicing procedures. These essential but costly items include robotic systems, laparoscopic tools, and other specialized instruments, which require substantial investments for maintenance and upgrades. In addition, operating rooms and simulation labs are major expenses. These facilities require surgical beds, instruments, and disposable materials. Finally, special case studies may create additional costs for real-world case studies, and clinical training also includes malpractice insurance for medical schools, amounting to tens of thousands of dollars annually.

¹¹ Hanson, Melanie, and Fact Checked. "Average Cost of Medical School [2024]: Yearly + Total Costs." *Education Data Initiative*, 9 Nov. 2024, educationdata.org/average-cost-of-medical-school. Accessed 19 Nov. 2024.

¹² "AAMC Statement on Health and Human Services Funding in FY 2024 Appropriations Legislation." AAMC, 25 Mar. 2024, www.aamc.org/news/press-releases/aamc-statement-health-and-human-services-funding-fy-2024-appropriations-legislation. Accessed 19 Nov. 2024.

Even more, the FY 2024 budget addresses the increasing cost of training in primary care medicine, for healthcare providers, and in research, amounting to approximately 1.1 billion dollars in just 2024 alone.¹³

	2022	2023	2024	2024+/-2023
Training in Primary Care Medicine	49	50	54	+4
Training for Health Care Providers	--	--	5	+5
Research Training	967	1,034	1,051	+17

Figure 1.3¹⁴: Costs of training(in millions of dollars) in 3 separate categories from 2022 to 2024

The budget also includes \$106 million allocated within the CDC to support public health training and fellowship programs. Furthermore, many training curricula and models for training health professionals, particularly in medicine, remain unchanged from decades ago—meaning most are not currently leveraging the available technological opportunities. Another \$28 million is planned for a new program to address growing concerns around healthcare workforce shortages, which will improve recruiting and training new providers.¹⁵

There are some existing solutions to these challenges, such as Osso VR and Touch Surgery, which offer immersive training for medical students and surgeons at reduced costs. One method used is tracking performance analytics which is then used to improve surgeon skills and pre-surgical planning, allowing for minimal errors during actual surgeries. Taking into account the hardware, software, training modules, customization, and integration, the costs would likely be around \$55,000 per simulated surgery, which is a fraction of traditional physical training costs. The use of VR solutions can provide a reusable and realistic anatomy model while providing scalable training experiences.

¹³ Assistant Secretary for Financial Resources (ASFR). “HHS FY 2024 Budget in Brief.” HHS.Gov, 16 Sept. 2024, www.hhs.gov/about/budget/fy2024/index.html. Accessed 19 Nov. 2024.

¹⁴ “President Biden Releases FY 2024 Budget Request.” AAMC, 10 Mar. 2023, www.aamc.org/advocacy-policy/washington-highlights/president-biden-releases-fy-2024-budget-request. Accessed 19 Nov. 2024.

¹⁵ “Health Care Workforce Innovation Program.” Energy Communities, 1 Apr. 2024, energycommunities.gov/funding-opportunity/health-care-workforce-innovation-program/. Accessed 19 Nov. 2024.

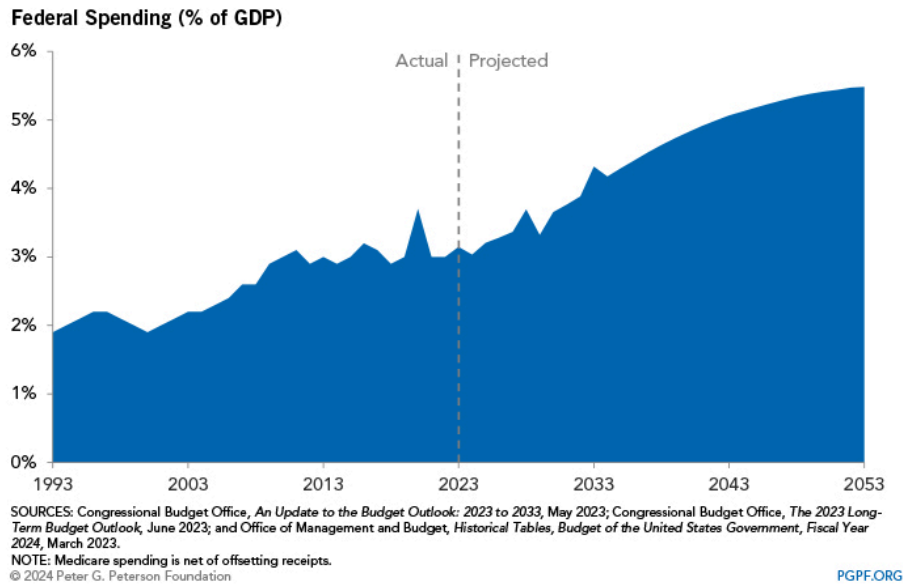


Figure 1.4¹⁶: Projected federal spending on healthcare.

These costs are reflected greatly in our economy, as shown above in Figure 1.4. The US government spent upwards of \$4.5 billion on healthcare in just 2022, and recent projections expect that cost to jump up by 8.9% in 2024 alone, and beyond, upwards of 6% of our GDP.^{17,18} Medical students today must take out loans to pay tuition fees just to live. Typical starting salaries are often not enough to cover the excess of student loan debt, often forcing physicians to only take high enough paying jobs, which causes medical prices to rise to cover the increase in medical tuition. In addition, the complexity of the US healthcare system leads to unwanted waste in insurance and provider payment systems, driving up federal spending. These challenges place additional unwanted pressure on healthcare professionals, reducing quality patient care and contributing to physician burnout.

¹⁶ “Historical.” CMS.Gov, www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data/historical. Accessed 12 Nov. 2024.

¹⁷ “Historical.” CMS.Gov, www.cms.gov/data-research/statistics-trends-and-reports/national-health-expenditure-data/historical. Accessed 12 Nov. 2024.

¹⁸ Mayer, Kathryn. “Insurers Say Health Care Costs Will Jump in 2024.” Welcome to SHRM, 20 May 2024, www.shrm.org/topics-tools/news/benefits-compensation/health-care-costs-2024-growth-wtw-buck. Accessed 12 Nov. 2024.

This situation of the high costs of obtaining a medical degree diminishes its value, costing (tuition fees, student loans) more than the benefits (future earning potential and career growth).¹⁹ This has led some to question whether the value of a medical degree is worth the financial burdens that follow. The rising pace of inflation also outpaces many starting salaries, deepening the risk of debts and financial struggle. Given these tendencies, newer solutions are being developed that offer cheaper alternatives to reduce the time and cost of medical training without sacrificing quality. A potential solution would be a haptic VR product that would create a new, risk-free, and life-like training environment that is also cost-effective in the long term.

One such study compared the cost impact of VR and live evacuation training. “The results showed that the initial costs for VR were higher than the live training, but within 3 years, the cost per participant for the VR training was much lower than the cost per patient for the live drill. The live exercises maintain a fixed cost, but after the initial costs of VR, the use of VR becomes less expensive.”²⁰

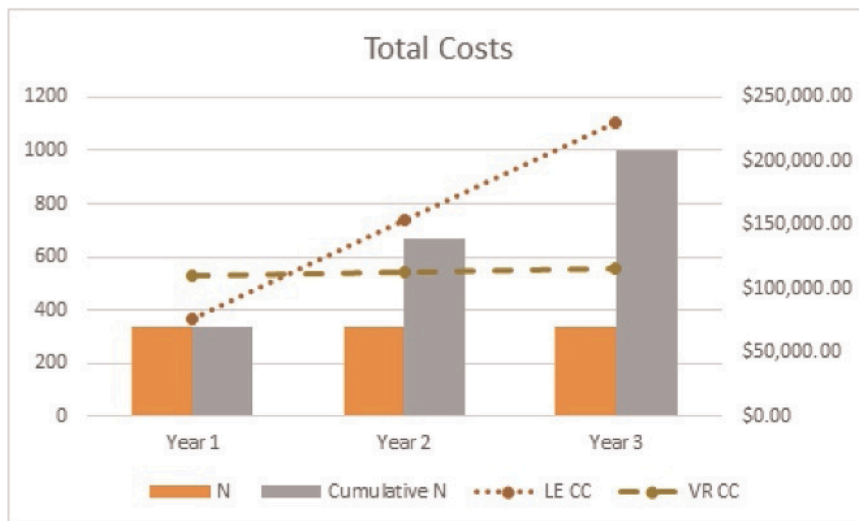


Figure 1.5²¹: Number of participants trained versus training costs.

This trend is visualized in Figure 1.5, showcasing increasing medical training costs versus digital alternatives. With its flexible customizations and lower long-term costs, VR enables crucial hands-on experience while helping to alleviate the thinning financial bubble.

¹⁹ Walsh, K. “The Cost Bubble in Medical Education: Will It Burst and When?” Annals of Medical and Health Sciences Research, U.S. National Library of Medicine, [pmc.ncbi.nlm.nih.gov/articles/PMC5405639/](https://pubmed.ncbi.nlm.nih.gov/articles/PMC5405639/). Accessed 12 Nov. 2024.

²⁰ Tmonroy. “The Top 8 Ways Virtual Reality in Healthcare Is Transforming Medicine.” XR Health, 2 Aug. 2024, www.xr.health/the-top-8-ways-virtual-reality-in-healthcare-is-transforming-medicine/. Accessed 14 Nov. 2024.

²¹ “Comparative Cost of Virtual Reality Training and Live Exercises for Training Hospital Workers for Evacuation.” Computers, Informatics, Nursing : CIN, U.S. National Library of Medicine. Accessed 14 Nov. 2024.

III. Market

A. Market Size

Across the country, different institutions and companies are working hard to train the current and next generation of medical professionals with the most up-to-date pieces of technology. According to the Alliance for Academic Internal Medicine (AAIM), in 2014 the total cost to train a medical resident for a single year was 183,416 dollars.²² Also according to the Association of American Medical Colleges, in 2024, there are currently 97,903 people currently enrolled in medical school.²³ These numbers together indicate that the medical training field is worth over 17 billion dollars.

B. Competition and Obstacles

A great portion of competition in the medical training field is large healthcare companies such as GE Healthcare, Johnson and Johnson, and Medtronic. However, the companies that are the biggest potential issues are those focused specifically on improvements within the medical field, such as Laerdal Medical, Touch Surgery, and Osso VR. Normally, the largest barrier to entering the medical training field is money, as surgical equipment and trainers are quite expensive. In addition to monetary challenges, federal approval of training techniques can also be a lengthy legal process.²⁴ Opportunities exist to create a product that will be able to subside the typical costs by creating a digital environment that alleviates the need for expensive machinery and operating costs, improving the efficiency of the medical training process, such that existing training paths can implement solutions without the need for lengthy legal work.

C. Consumers in the Medical Training Field

The COVID-19 pandemic sparked a lot of change within both the world and the world's mindsets. People saw a need for more medical professionals which caused close to a 20% increase in the number of applications sent to medical schools in 2021.²⁵ Following the pandemic, the number

²² Allen, James, et al. "The Total Cost To Train A Physician." *What I've Learned As A Hospital Medical Director*, 11 July 2019, hospitalmedicaldirector.com/the-total-cost-to-train-a-physician/. Accessed 14 Nov. 2024.

²³ "2023 FACTS: Enrollment, Graduates, and MD-PhD Data." AAMC, www.aamc.org/data-reports/students-residents/data/2023-facts-enrollment-graduates-and-md-phd-data. Accessed 14 Nov. 2024.

²⁴ Turinas, Adam. "8 Most Common Healthcare Market Entry Barriers & How To Overcome Them." Health Launchpad, 4 June 2024, healthlaunchpad.com/8-most-common-healthcare-market-entry-barriers-how-to-overcome-them/. Accessed 14 Nov. 2024.

²⁵ Colettehanngo. "Medical Students Haven't Been Deterred by the Stress of Covid on Hospitals and Staff - They Are Eager to Help." *CNBC*, 6 Apr. 2021, <https://www.cnbc.com/2021/04/06/applications-to-medical-schools-surged-during-the-covid-pandemic.html>. Accessed 14 Nov. 2024.

continues to increase, over the past 4 years there has been an increase of 5000 students enrolled in medical school.²⁶

However, while COVID did lead to a sudden surge in this number, the number of medical students has been steady for the past two decades.

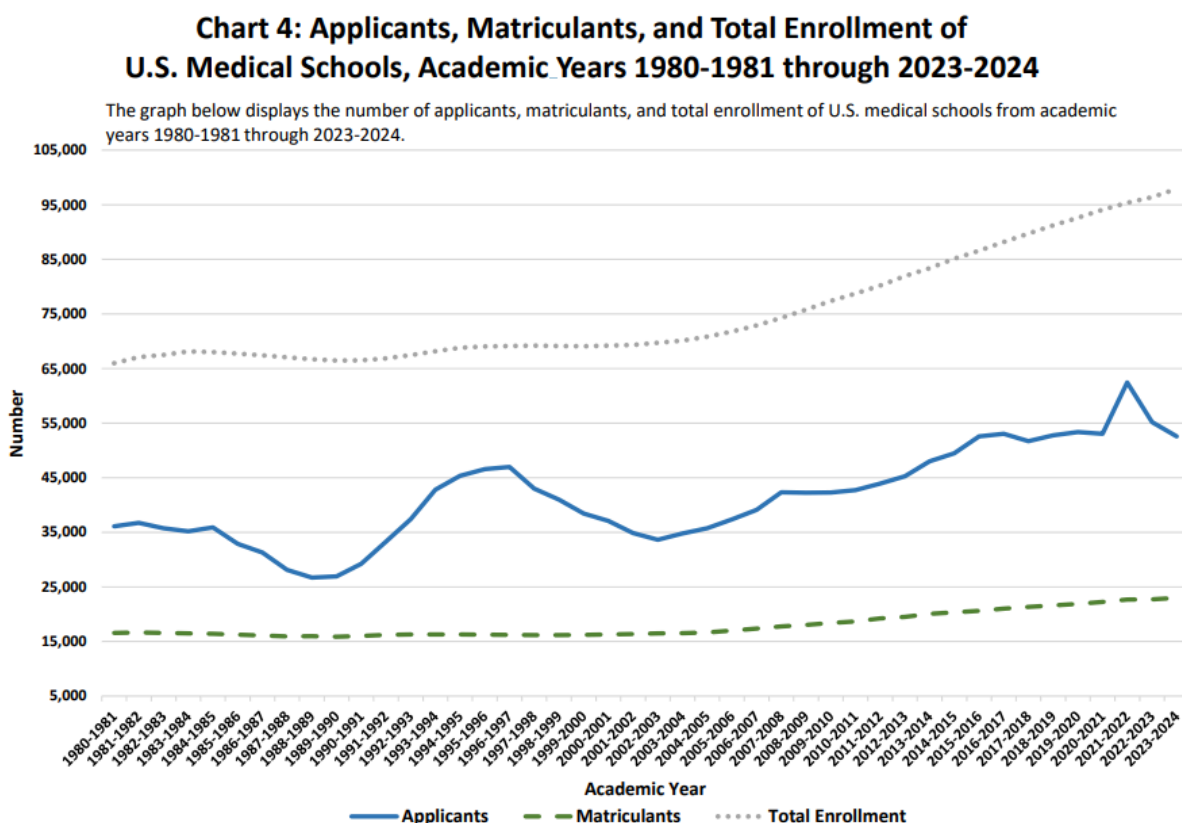


Figure 1.6²⁷: Graph showing the relationship between number of Medical School Applications, Matriculants, and Total Enrollment between 1980 and 2024.

As Figure 1.6 shows, before the year 2000, the number of medical students remained fairly stagnant, however, over the past two decades the number of medical students has been increasing by around a thousand each year. Due to the rising number of medical students, Colleges and Universities are looking for easier and more effective training methods to keep up with enrollment. Furthermore, marketing toward universities should prove beneficial as Institutions across the United States spend over 30 billion dollars annually on R&D within the medical field alone, which is growing. From 2010 to 2022 this amount rose by 57%.²⁸

²⁶ “2023 FACTS: Enrollment, Graduates, and MD-PhD Data.” AAMC, <https://www.aamc.org/data-reports/students-residents/data/2023-facts-enrollment-graduates-and-md-phd-data>. Accessed 14 Nov. 2024.

²⁷ “Chart 4: Applicants, Matriculants, and Total Enrollment of U.S. Medical Schools.” AAMC, 11 6 2023, <https://www.aamc.org/media/37836/download>. Accessed 17 11 2024.

²⁸ “Higher Education Research and Development (HERD) Survey 2022 | NSF - National Science Foundation.” *National Center for Science and Engineering Statistics*, <https://nces.nsf.gov/surveys/higher-education-research-development/2022#data>. Accessed 13 November 2024.

D. Survey Results

A survey focusing on medical field training, virtual reality, and its potential incorporation within the medical field was sent out to current medical students, recent graduates, and current medical professionals. The following are the results of the 33 respondents who answered the survey.

On a scale from 1 to 10, do you believe there is a lack of hands-on resources and practice for physical procedures and surgeries in the medical field?

33 responses

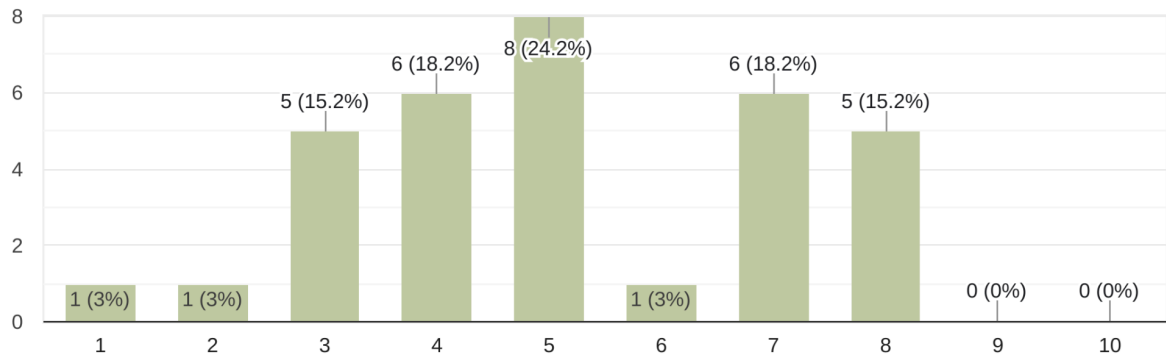


Figure 1.7: Survey results for the question asking for participants' beliefs on if there is/was a lack of hands-on practice for surgeries and other physical operations.

Figure 1.7 shows that surgeons are indifferent on whether or not there is a lack of hands-on preparation for their jobs. The average rating out of 10 was 5.15, which means there is neither an abundance nor an absence of training for these kinds of operations.

On a scale from 1 to 10, how effective do you believe VR could be in enhancing medical training and preparation?

33 responses

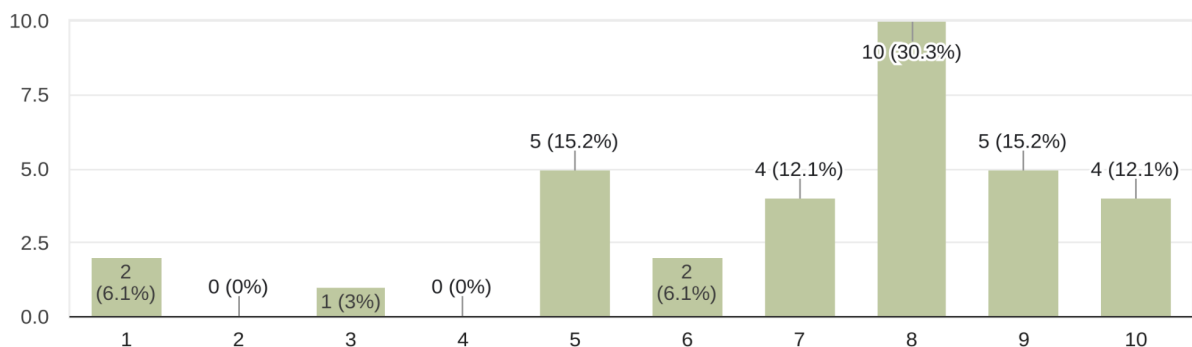


Figure 1.8: Survey results for the question about whether virtual reality could have the potential to improve medical training.

Fig. 1.8 shows how people feel, on a scale of one to ten, about whether virtual reality would have the ability to improve medical training and preparation for performing surgeries and other medical procedures. The average response was 7.12 out of 10, meaning that medical professionals do believe that VR can have applications within the medical training field, however, a few were strongly against it.

A follow-up question was asked about the reasoning behind the respondent's choices. Those who responded positively towards the idea of VR in the medical field mentioned its safety benefits and its ability to learn through doing as opposed to lecture-style learning.

On a scale from 1 to 10, how important do you consider incorporating touch sensation (haptic feedback) for effective VR-based medical training?
33 responses

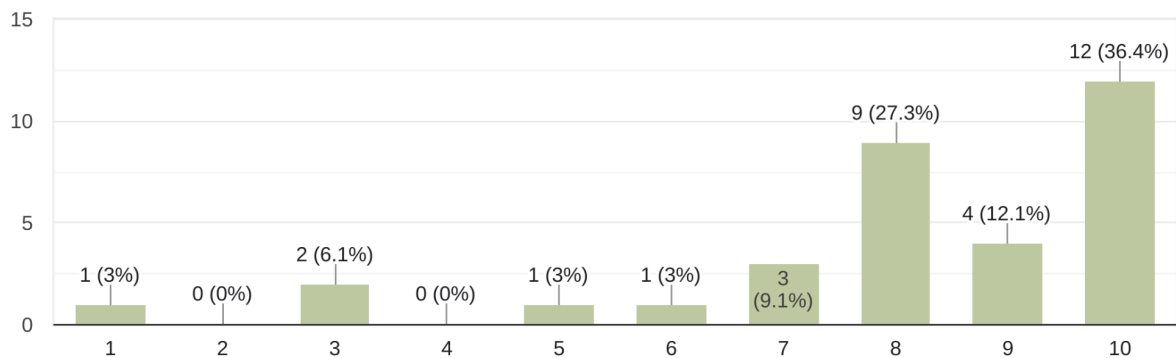


Figure 1.9: Survey results for the question about how important haptic feedback hands would be for effective VR medical training.

Fig. 1.9 shows how important the incorporation of haptic feedback would be for virtual reality to be an effective training tool for medical operations. The average response was 8.09 out of 10, showing that medical professionals strongly believe that for virtual reality to be an effective tool in medical training, a form of force feedback is strongly needed. A follow-up question was once more asked about the reason for their answers, and almost every single respondent mentioned that to develop safe and correct habits during surgery, trainees need to be able to feel how much force is required for certain movements, and mentioned that the precision that could be achieved with a better user interface could substantially enhance learning opportunities.

IV. Conclusion

Rising medical costs pose significant challenges for future patients and students, while traditional training methods like cadavers face limitations such as scarcity, high costs, and ethical concerns. As the financial and logistical burdens of medical education grow, virtual reality offers an innovative solution with immersive haptic feedback, creating a cost-effective, efficient, and risk-free training environment. VR technology has the power to not only reduce costs and enhance learning outcomes, but also pave the way for a future of new medical training and patient care.